Optimizing Foundation Models

Introduction

Two techniques to improve the performance of a foundation model (FM):

- Retrieval Augmented Generation (RAG)
- Fine-tuning.

Objectives

- Identify AWS services that help store embeddings with vector databases.
- Understand the role of agents in multi-step tasks.
- Understand approaches to evaluate FM performance.
- Determine whether an FM effectively meets business objectives.
- Define methods for fine-tuning an FM.
- Describe how to prepare data to fine-tune an FM.
- Determine whether an FM effectively meets the business objectives based on the business metric identified in the use case.

Introduction to AnyCompany, a telecom company business case study

Business Case 1 – Read this use case in the Word document: AWS AI Practitioner Class Document.docx

After the Business Case reading....

In the next lesson, you will learn about how RAG can help AnyCompany.

Retrieval-Augmented Generation

From Dataset to Vector Embeddings

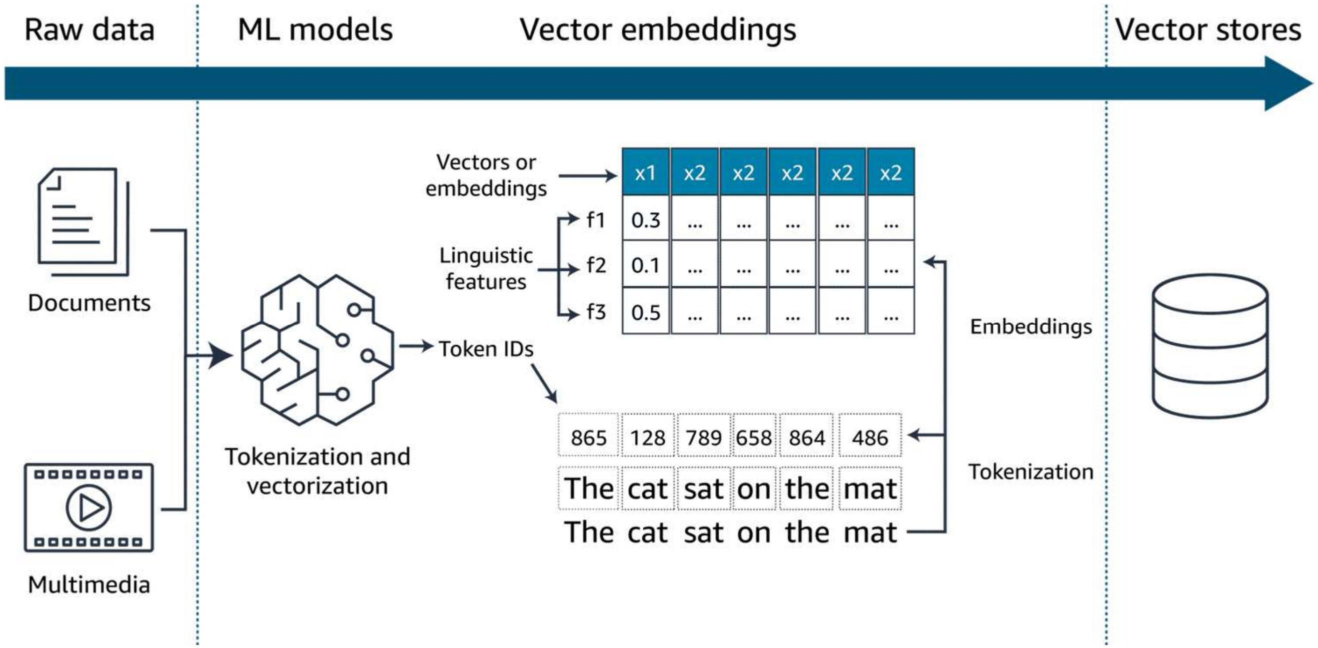
Enterprise Datasets and LLMs:

- While LLMs are powerful, they often lack the domain-specific knowledge needed for enterprise use without access to internal data.
- Enterprises store large volumes of internal documents, reports, and manuals that are typically unknown to foundation models.
- Supplying relevant enterprise data as context alongside prompts helps LLMs generate more accurate, tailored outputs.
- Vector embeddings are used to search and retrieve the most relevant context from enterprise datasets to enhance the model's responses.

Retrieval-Augmented Generation

Vector Embedding is the process by which text, images and audio are given numerical representation in a vector space. Embedding is usually performed by a machine learning (ML) model.

Enterprise datasets, such as documents, images and audio are passed to ML models as tokens and are vectorized. These vectors in an n-dimensional space, along with the metadata about them, are stored in purpose-built vector databases for faster retrieval.



Retrieval-Augmented Generation

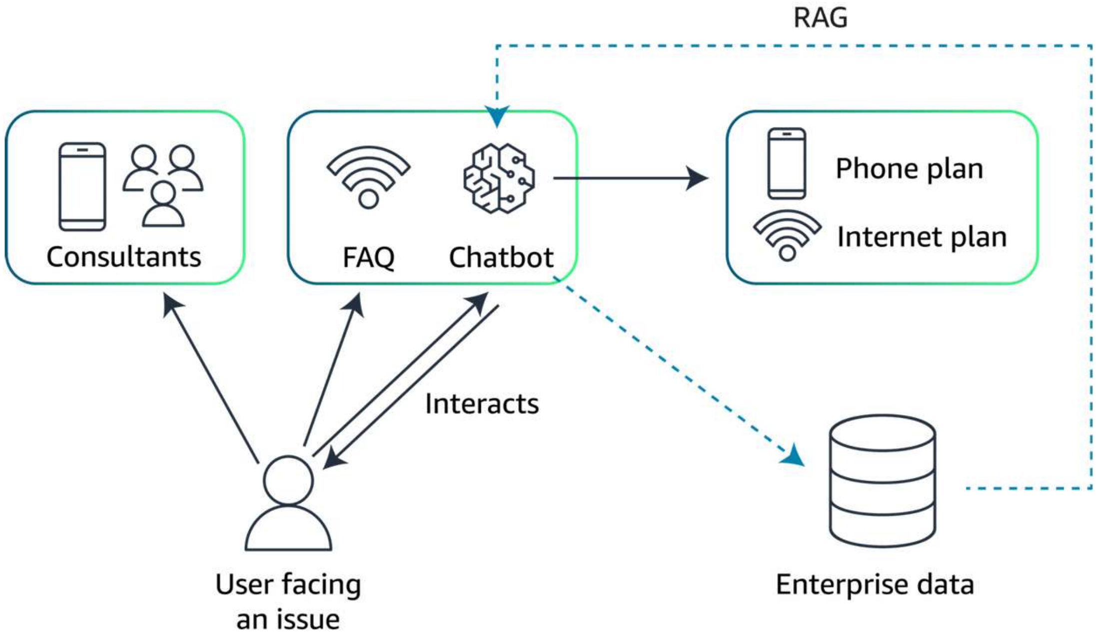
Storing Vectors

Vector databases provide ultra-fast similarity searches across these billions of vectors in real time.

Amazon Web Services (AWS) offers the following viable vector database options:

- **Amazon OpenSearch Service** (provisioned)
- Amazon OpenSearch Serverless
- pgvector extension in **Amazon Relational Database Service (Amazon RDS) for PostgreSQL**
- pgvector extension in **Amazon Aurora PostgreSQL-Compatible Edition**
- **Amazon Kendra**

Retrieval-Augmented Generation



An updated version of AnyCompany's architecture diagram, including the RAG system. The chatbot is now able to query a database containing enterprise data and use it to provide more accurate and contextual answers to users.

Agents

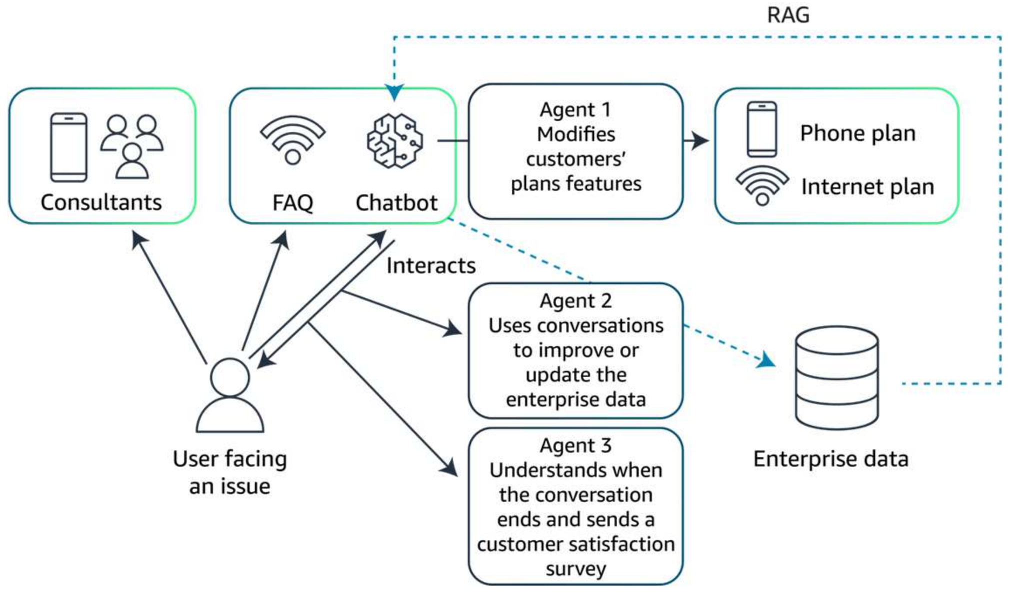
Roles of Agents in Generative AI Applications:

- **Intermediary Operations:** Agents bridge generative AI models with backend systems like databases, CRMs, or service tools, enabling smooth interaction and data flow.
- **Action Execution:** They perform tasks such as adjusting settings, processing transactions, or retrieving documents based on user intent interpreted by the AI.
- **Feedback Integration:** Agents gather outcome data from their actions, which helps improve the AI model's accuracy and performance over time.

Agents

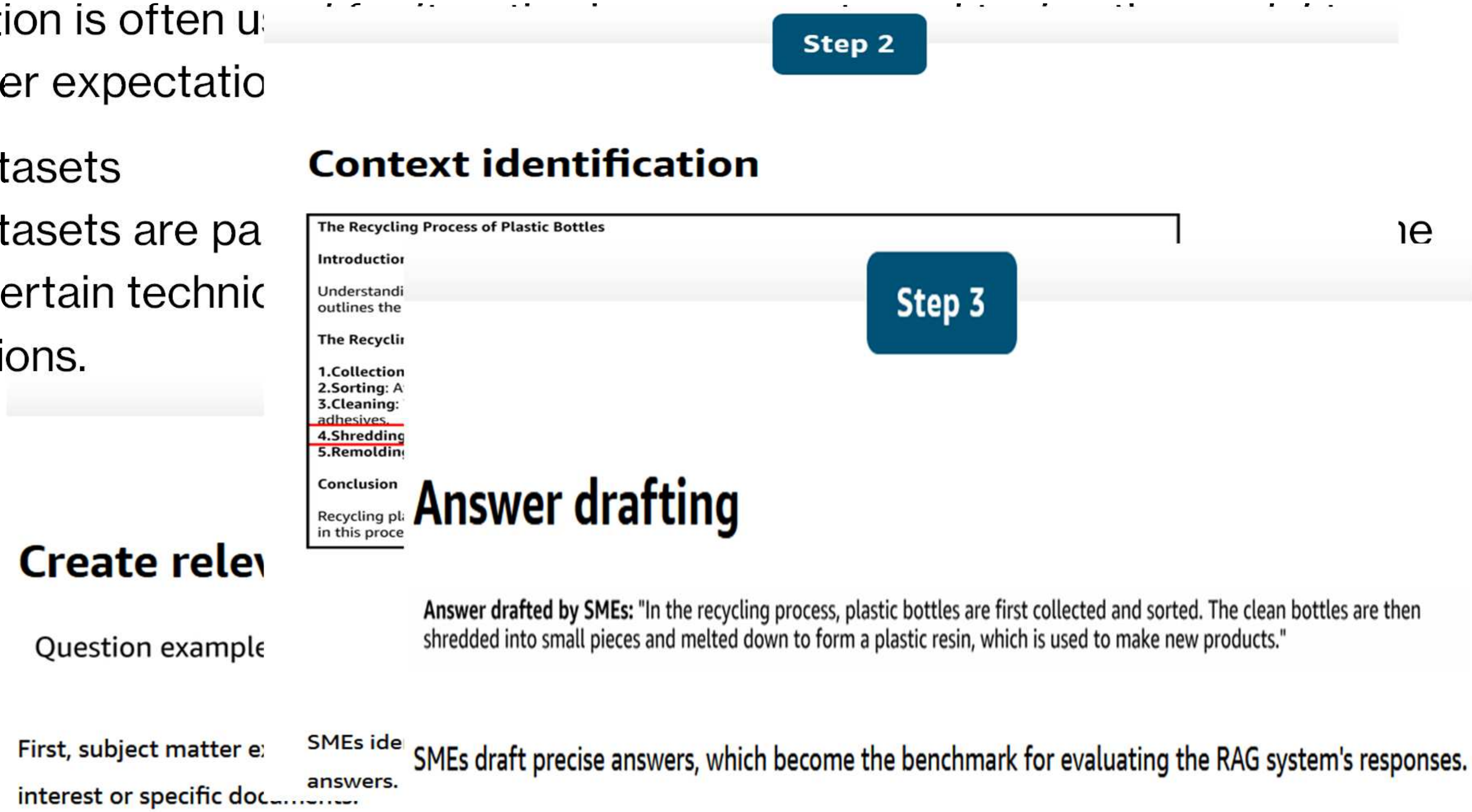
AnyCompany's Agent-Based Architecture (Detailed View):

- **Agent 1** handles user prompts related to customer plans by taking specific actions based on those prompts.
- **Agent 2** updates or enriches the enterprise database using information from the user-chatbot conversation, enhancing future responses via the RAG system.
- **Agent 3** sends a satisfaction survey at the end of the conversation, helping track and improve customer satisfaction scores.



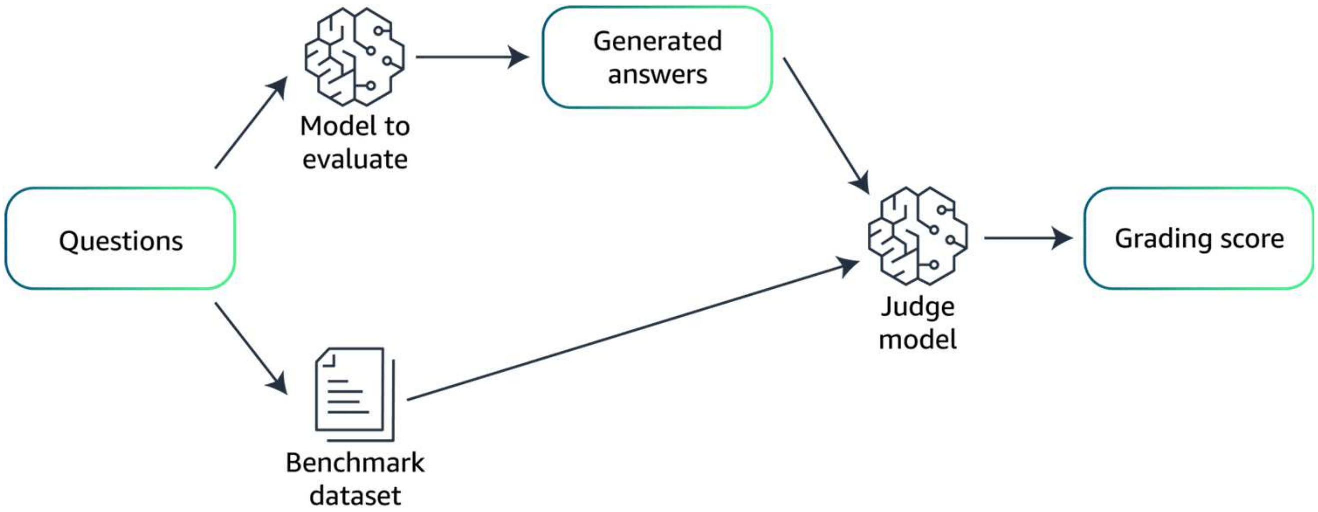
Evaluate Results

- Human evaluation
  Human evaluation is often u
  better meet user expectatio
- Benchmark datasets
  Benchmark datasets are pa ... ne
  model meets certain technic
  human evaluations.



Evaluate Results

The evaluation of LLM performance using a benchmark dataset can be automated using an *LLM as a judge* approach.



Optimizing Foundational Model with Fine-tuning

Business Case 2 – Read this use case in the Word document: AWS AI Practitioner Class Document.docx

AnyCompany's LLM-Based Solution:

- **Targeted Data Utilization:** Fine-tuned on transactional records, customer feedback, and interaction data to enhance personalization.
- **Real-Time Personalization:** Integrated with a recommendation engine to dynamically tailor product displays and promotions for each customer.
- **Continuous Learning:** The model self-updates with new customer data and fashion trends to maintain accuracy without manual reconfiguration.